



UNIVERSITY OF PISA

Master's Degree in Digital Humanities

## Data Mining: Fundamentals

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# Chapter 1: Data Understanding and Preparation

This chapter introduces the meaning and characteristics of the variables in the dataset used for the subsequent analyses. The dataset contains information on more than 20,000 board games sourced from the BoardGameGeek platform: each instance represents a single game, while the variables describe its structural characteristics, the ratings provided by the community, popularity indicators, and classifications across different thematic categories. Below is Table 1.1, which reports the variables with the characteristics just described.

| Variables                                                                                        | Types                         | Description                                                                     |
|--------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------|
| BGGId                                                                                            | Numeric                       | Unique identifier of the game on the platform                                   |
| Name                                                                                             | Categorical (string)          | Name of the game                                                                |
| Description                                                                                      | Textual                       | Game description entered by users                                               |
| YearPublished                                                                                    | Numeric                       | Year the game was published                                                     |
| GameWeight                                                                                       | Numeric                       | Difficulty estimated by users (1–5)                                             |
| ComWeight                                                                                        | Numeric                       | Difficulty recommended by the community (1–5)                                   |
| MinPlayers                                                                                       | Numeric                       | Minimum number of players                                                       |
| MaxPlayers                                                                                       | Numeric                       | Maximum number of players                                                       |
| ComAgeRec                                                                                        | Numeric                       | Minimum age recommended by the community                                        |
| LanguageEase                                                                                     | Numeric                       | Language dependence (need to read / rule complexity)                            |
| BestPlayers                                                                                      | Categorical                   | Ideal number of players according to the community                              |
| GoodPlayers                                                                                      | Categorical (string list)     | List of recommended player counts                                               |
| NumOwned                                                                                         | Numeric                       | Number of users who own the game                                                |
| NumWant                                                                                          | Numeric                       | Number of users who want the game                                               |
| NumWish                                                                                          | Numeric                       | Number of users who have the game in their wishlist                             |
| NumWeightVotes                                                                                   | Numeric                       | Total number of votes received on difficulty                                    |
| MfgPlayTime                                                                                      | Numeric                       | Duration estimated by the manufacturer                                          |
| ComMinPlaytime                                                                                   | Numeric                       | Minimum duration estimated by the community                                     |
| ComMaxPlaytime                                                                                   | Numeric                       | Maximum duration estimated by the community                                     |
| MfgAgeRec                                                                                        | Numeric                       | Minimum age recommended by the manufacturer                                     |
| NumUserRatings                                                                                   | Numeric                       | Total number of user ratings                                                    |
| NumComments                                                                                      | Numeric                       | Total number of user comments                                                   |
| NumAlternates                                                                                    | Numeric                       | Number of alternative versions                                                  |
| NumExpansions                                                                                    | Numeric                       | Number of linked expansions                                                     |
| NumImplementations                                                                               | Numeric                       | Number of games implementing the same system                                    |
| IsReimplementation                                                                               | Binary                        | Indicates whether the game is a reimplementation                                |
| Family                                                                                           | Categorical                   | Series or family the game belongs to                                            |
| Kickstarted                                                                                      | Binary                        | Indicates whether the game was funded via Kickstarter                           |
| ImagePath                                                                                        | Textual                       | URL of the game image                                                           |
| Rank:strategygames, Rank:abstracts, Rank:familygames                                             | Numeric                       | Rankings in the Strategy, Abstract, and Family categories.                      |
| Cat:Thematic, Cat:Strategy, Cat:War, Cat:Family, Cat:CGS, Cat:Abstract, Cat:Party, Cat:Childrens | Binary                        | Category membership flags (1 if the game belongs to the category, 0 otherwise). |
| Rating                                                                                           | Categorical (low/medium/high) | Overall rating used as the target variable                                      |

Table 1.1: Table 1 lists the complete set of available variables, along with their corresponding type and a brief description. Rank and category features are grouped.

## 1.1 DATA SEMANTIC

This section examines the most significant variables in the dataset, with particular attention to their semantic meaning in the context of board games. In general, a game

can be described through intrinsic attributes (such as year of publication, minimum and maximum number of players, estimated playing time, recommended minimum age, and complexity level), as well as through attributes derived from community interaction (difficulty ratings, number of owners, wishlist counts, and rankings within thematic categories). Some variables also provide information on structural relationships among games, such as the presence of expansions, implementations, or reimplementations.

- GameWeight and ComWeight represent the level of game complexity as perceived by users of the platform. Both variables take values ranging from 1 to 5, with higher values indicating games perceived as more complex. GameWeight corresponds to the average of the received ratings, whereas ComWeight is a more stable estimate obtained by taking into account the entire distribution of votes.
- LanguageEase indicates the degree of a game’s reliance on written language, based on community ratings. Higher values generally correspond to a greater amount of text or to the need to consult linguistic elements during gameplay.
- BestPlayers and GoodPlayers summarize community preferences with respect to the optimal number of players. BestPlayers identifies the number considered ideal, whereas GoodPlayers reports a set of values regarded as suitable
- NumOwned, NumWant, and NumWish are indicators of popularity on BoardGameGeek: NumOwned denotes the number of users who own the game, NumWant the number of users who would like to purchase it, and NumWish the number of users who have added it to their wishlist. These variables reflect the level of interest in a given title.
- Ranking variables indicate the position of a game within the platform’s different thematic rankings (Strategy, Family, Abstract, Thematic, etc.). These are ordinal variables: lower values correspond to better positions within the reference
- Variables describing relationships between games, such as NumExpansions, NumAlternates, NumImplementations, and IsReimplementation, highlight connections among titles, for example the presence of expansions or derivation from pre-existing games. These variables provide useful information for understanding the structure and evolution of game families.
- Finally, the Rating variable, divided into three levels (low, medium, and high), represents the target variable of the classification task defined in the project and summarizes the overall evaluation expressed by users.

## 1.2 DISTRIBUTION OF THE VARIABLES AND STATISTICS

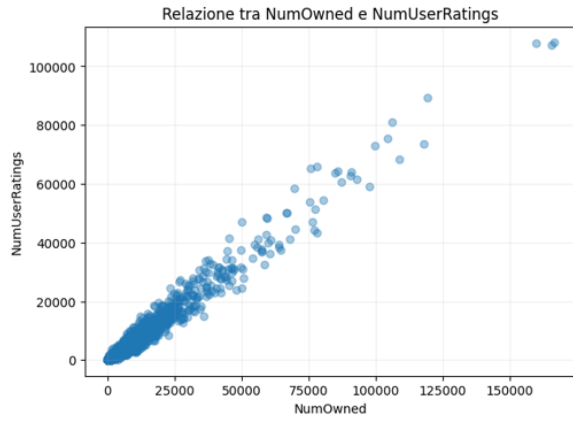
For an initial quantitative analysis, descriptive statistics of the main numerical variables were computed and are reported in Table X.1. These statistics highlight the strong skewness of several popularity-related variables (NumOwned, NumUserRatings, NumWish), as well as the presence of outliers in YearPublished, which will be addressed during the Data Quality phase. To avoid presenting an excessive number of plots, the distributions of all individual variables are not shown; instead,

Figure(1.1) presents a selection of representative graphs that emerged from the exploratory analysis.

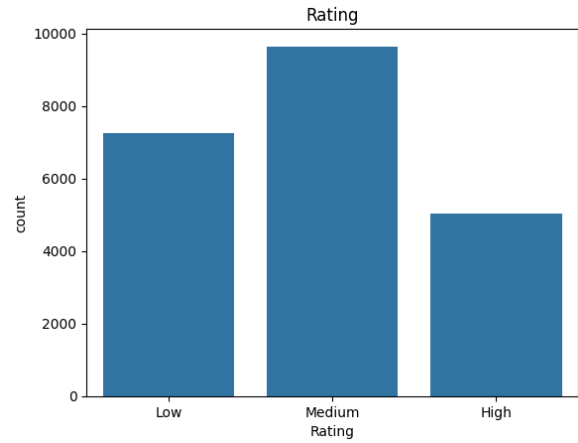
- ComWeight: Game complexity, as rated by users, exhibits a right-skewed distribution: the majority of titles are concentrated around low to medium values rated by 2 (1–3), while, while highly complex games are less frequent. This suggests that the dataset is dominated by games of moderate complexity, whereas so-called *heavy games* represent a smaller portion of the sample.
- NumUserRatings ( $\log_{10}$ ): The number of user ratings exhibits a long-tail distribution; to improve readability, a  $\log_{10}$  transformation was applied. Even after the transformation, strong heterogeneity can be observed: a small number of titles receive a very large number of reviews, while most games accumulate a limited number of ratings.
- YearPublished: Grouping games by time intervals highlights a strong increase in production during the 2010–2019, decade, consistent with the recent growth of the industry. Some implausible values are present in the dataset and will be addressed during the Data Quality phase by restricting the analysis to a plausible time range (1850–2025).
- Rating: The target variable is imbalanced: the Medium category is the most frequent, while Low and High are less represented.
- Relationship between popularity measures. The scatter plot of NumOwned versus NumUserRatings Figure(1.2) shows a clear positive relationship: games owned by a larger number of users also tend to receive more ratings. This confirms the consistency between popularity measures already observed in the correlation values and in the patterns reported in Table 1.1.

| Variable        | Mean   | Std     | Min   | Max    | Skewness |
|-----------------|--------|---------|-------|--------|----------|
| ComWeight       | 2,18   | 0,86    | 0     | 5,21   | 0,30     |
| NumUserRatings  | 861,67 | 3638,68 | 30    | 108101 | 12,59    |
| NumWish         | 228,46 | 788,48  | 0     | 19182  | 9,35     |
| YearPublished   | 1985,5 | 212,5   | -3500 | 2021   | -11,32   |
| Ratings (count) | –      | –       | –     | –      | –        |

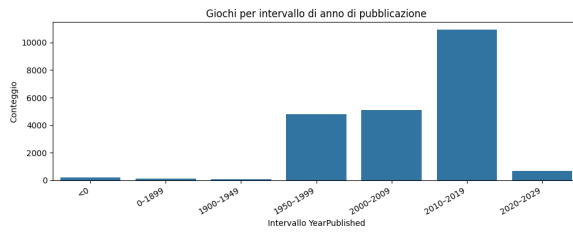
Table 1.2: Descriptive statistics of the main numerical variables in the dataset



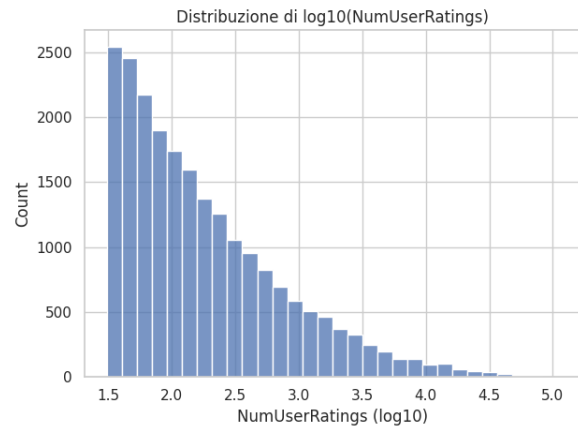
(a) Scatter Plot: Relationship NumOwned NumUserRatings



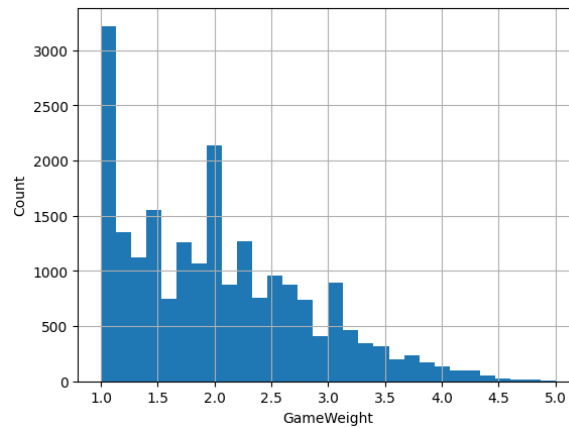
(b) Rating Bar Chart



(c) Publication Year Range



(d) NumUserRatings Distribution



(e) GameWeight Histogram

Figure 1.1: Data representation

## 1.3 DATA QUALITY

### 1.3.1 ERRORS

Several values violating the semantic consistency of the dataset variables were identified. These values were not manually corrected, but were instead marked as missing

values (NaN), with their handling deferred to the data preparation phase. The identified errors fall into the following categories:

- Out-of-domain values: The variable *ComWeight*, which represents a measure of game complexity on a  $[1,5]$  scale, exhibits 519 observations outside this range. These values were considered semantically invalid and were marked as missing.
- Logical inconsistencies between related variables: A total of 182 observations were identified in which *MinPlayers* or *MaxPlayers* assume zero or negative values, or where  $\text{MaxPlayers} < \text{MinPlayers}$ . These observations were marked as missing and subsequently removed.
- Non-plausible temporal values: The variable *YearPublished* contains 274 values outside the plausibly defined interval  $[1850,2025]$ , which were marked as missing.
- Inconsistencies in playtime variables: For the variables *ComMinPlaytime* and *ComMaxPlaytime*, 151 observations were identified with negative values or with  $\text{ComMaxPlaytime} < \text{ComMinPlaytime}$ . These observations were also considered semantically invalid and converted into missing values.
- Impossible numerical values: For some count variables (e.g., *NumOwned*, *NumUserRatings*, *NumWish*), negative values were detected and marked as missing.
- Duplicates: A duplicate check was performed using *BGGId* as the unique game identifier. The analysis did not reveal the presence of duplicate records in the dataset.

### 1.3.2 MISSING VALUES

The analysis of missing values was carried out in two phases. First, the missing values originally present in the dataset at the time of data loading were identified. Subsequently, additional missing values were introduced during the semantic validation phase, when inconsistent or implausible values were marked as missing, as described in the previous section. After the semantic validation phase, missing values were handled as follows:

- for continuous numerical variables (*ComAgeRec*, *YearPublished*), median imputation was applied, as it is robust to the presence of outliers;
- for *ComWeight*, as an ordinal variable, the use of the median is methodologically appropriate and preserves the consistency of the scale;
- Observations with missing *MinPlayers* and *MaxPlayers* values were removed, as they correspond to semantic errors that cannot be reliably imputed;
- Observations with missing *ComMinPlayTime* and *ComMaxPlayTime* were also removed.
- The *LanguageEase* variable was not imputed, as it contains structural missing values, and was subsequently excluded from the predictive models;
- the remaining variables with residual NaN values do not affect the subsequent analyses, as they will be removed during the non-informative feature selection phase.

## 1.4 OUTLIERS AND VARIABLES TRANSFORMATION

Outlier analysis was performed on the numerical variables of the dataset, with particular attention to those related to game popularity, namely NumOwned, NumUserRatings, and NumWish. These variables exhibit highly skewed distributions, characterized by long tails and the presence of extreme values.

Outliers were identified using boxplots (Figure 1.2), which highlight the presence of observations far removed from the central mass of the data. However, these extreme values do not correspond to measurement errors or noise; rather, they represent games that are particularly popular on the platform.

For this reason, outliers were not removed, as they carry relevant information and are representative of the phenomenon under analysis. Their removal would have resulted in a distortion of the original data distribution and a significant loss of information.

In order to reduce the effect of strong skewness and improve the readability of the distributions, a logarithmic transformation was applied to selected variables. In particular, the transformations  $\log_{10}(\text{NumUserRatings} + 1)$  and  $\log_{10}(\text{NumWish} + 1)$  were considered, as they mitigate the impact of extreme values while preserving the relative ordering of observations and allowing the inclusion of cases with zero values.

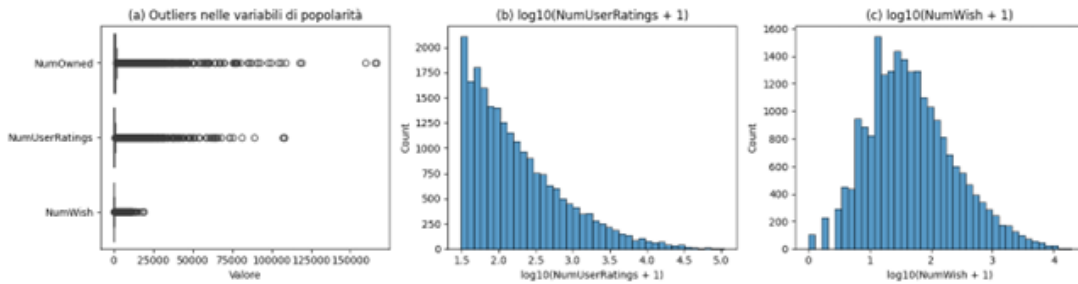


Figure 1.2: Outliers Analysis

## 1.5 CORRELATIONS AND ELIMINATION OF VARIABLES

|                    | YearPublished | ComWeight | MinPlayers | MaxPlayers | ComGenre  | LanguageEn | BestPlayers | NumWish   | ComMinPlay | ComMaxPlay | MinGenre  | NumUserRatings | NumReviews | NumQuestions | NumImplementations | TabImplementations | Kickstarted | Cat:Thematic | Cat:Strategy | Cat:War   | Cat:Family | Cat:OS    | Cat:Party | Cat:Childrens |
|--------------------|---------------|-----------|------------|------------|-----------|------------|-------------|-----------|------------|------------|-----------|----------------|------------|--------------|--------------------|--------------------|-------------|--------------|--------------|-----------|------------|-----------|-----------|---------------|
| YearPublished      | 1.00000       | 0.040500  | -0.00482   | -0.00343   | 0.050935  | 0.020115   | 0.010202    | 0.020158  | 0.004683   | 0.000880   | 0.003121  | 0.000939       | -0.119588  | 0.017095     | 0.019003           | 0.030883           | 0.027623    | 0.024078     | 0.035100     | 0.021242  | 0.019435   | 0.010884  | 0.021100  | -0.028853     |
| ComWeight          | 0.040500      | 1.00000   | -0.171861  | -0.089443  | 0.048016  | -0.008411  | 0.147885    | 0.180896  | 0.128581   | 0.181304   | 0.281312  | 0.080402       | -0.038108  | 0.128047     | 0.030927           | 0.040008           | 0.060793    | 0.128238     | 0.290435     | 0.444825  | -0.128081  | 0.096384  | -0.128837 | -0.184887     |
| MinPlayers         | -0.00482      | -0.171861 | 1.00000    | 0.834351   | -0.022436 | -0.003880  | 0.017088    | -0.078881 | 0.043088   | 0.014068   | 0.037847  | -0.012331      | 0.004208   | -0.041388    | 0.018881           | -0.008807          | -0.123042   | -0.077102    | -0.008895    | -0.141236 | 0.001089   | -0.018910 | 0.108762  | -0.013228     |
| MaxPlayers         | -0.00343      | -0.089443 | 0.834351   | 1.00000    | -0.011184 | -0.010216  | 0.002095    | -0.017845 | -0.008808  | -0.008411  | 0.010181  | -0.004469      | 0.009963   | 0.010185     | 0.018839           | 0.000141           | -0.018981   | -0.080794    | -0.024792    | -0.004788 | -0.001089  | -0.010481 | 0.088352  | -0.008762     |
| ComGenre           | 0.050935      | 0.048016  | -0.022436  | -0.011184  | 1.00000   | 0.010035   | 0.042855    | 0.070287  | 0.004208   | 0.013446   | 0.032816  | 0.003469       | -0.170055  | 0.008231     | -0.008370          | 0.000715           | 0.008540    | 0.010378     | 0.152068     | 0.016889  | -0.017761  | 0.033302  | 0.020813  | -0.246796     |
| LanguageEn         | 0.020115      | -0.008411 | -0.003880  | -0.010216  | 0.010035  | 1.00000    | -0.018181   | -0.140286 | 0.00812    | 0.004546   | 0.009529  | -0.139418      | -0.009072  | -0.047271    | -0.002908          | -0.000809          | -0.012858   | -0.048823    | -0.080405    | 0.028224  | -0.000380  | -0.000320 | 0.025108  | -0.031003     |
| BestPlayers        | 0.010202      | 0.147885  | 0.017088   | 0.002095   | 0.042855  | -0.018181  | 1.00000     | 0.500981  | -0.008808  | 0.008888   | 0.0114553 | 0.017888       | 0.018854   | 0.019199     | 0.219438           | 0.081300           | 0.032489    | 0.169685     | 0.380778     | -0.088121 | 0.188787   | -0.003804 | 0.117478  | -0.009700     |
| NumWish            | 0.020158      | 0.180896  | -0.078881  | -0.010485  | 0.070287  | -0.140286  | 0.500981    | 1.00000   | 0.010909   | 0.008551   | 0.013011  | 0.014345       | 0.081759   | 0.088177     | 0.174379           | 0.084002           | 0.070483    | 0.178722     | 0.310871     | -0.002114 | 0.114228   | 0.000084  | 0.028823  | -0.044095     |
| ComMinPlay         | 0.004683      | 0.128581  | 0.043088   | -0.008808  | 0.004208  | 0.00812    | -0.008808   | 0.001888  | 1.00000    | 0.834618   | 0.034883  | -0.004488      | -0.008931  | 0.001018     | 0.011438           | 0.002249           | -0.018181   | 0.004421     | 0.008158     | 0.118084  | -0.022827  | -0.005438 | -0.013117 | -0.018273     |
| ComMaxPlay         | 0.000880      | 0.181304  | 0.014068   | -0.008411  | 0.134046  | 0.004546   | 0.008888    | 0.008551  | 0.834618   | 1.00000    | 0.041280  | -0.004383      | -0.013083  | 0.004085     | 0.018857           | 0.013732           | -0.012858   | 0.003011     | 0.001153     | 0.018195  | -0.032700  | -0.000181 | -0.017085 | -0.028173     |
| MinGenre           | 0.003121      | 0.281312  | 0.037847   | 0.010181   | 0.032816  | 0.00812    | 0.114553    | 0.130101  | 0.034883   | 0.041280   | 1.00000   | 0.071329       | -0.004805  | 0.078282     | 0.008288           | 0.018042           | 0.088888    | 0.131870     | 0.178888     | 0.088088  | -0.004428  | 0.022009  | 0.022345  | -0.227331     |
| NumUserRatings     | 0.000939      | 0.080402  | -0.012331  | -0.004208  | 0.003469  | -0.139418  | 0.017888    | 0.014345  | -0.004488  | -0.004383  | 0.071329  | 1.00000        | 0.144837   | 0.000005     | 0.000020           | 0.047041           | 0.010780    | 0.006555     | 0.222395     | -0.008895 | 0.172887   | 0.018723  | 0.078802  | -0.024684     |
| NumReviews         | -0.119588     | -0.038108 | 0.004208   | 0.008808   | -0.170055 | -0.009072  | 0.118854    | 0.081759  | -0.008931  | -0.013083  | -0.004805 | 0.144837       | 1.00000    | 0.027595     | 0.001437           | 0.001383           | -0.041838   | 0.005380     | 0.004403     | -0.058144 | 0.003389   | -0.001883 | 0.048582  | 0.147485      |
| NumQuestions       | 0.017095      | 0.128047  | -0.008807  | -0.018981  | 0.008231  | 0.010181   | 0.004469    | 0.070287  | 0.008551   | 0.013011   | 0.014345  | 0.081759       | 0.088177   | 1.00000      | 0.027595           | 0.011844           | 0.081188    | 0.009993     | 0.127480     | 0.004407  | 0.007801   | 0.000990  | 0.116287  | 0.022886      |
| NumImplementations | 0.019003      | 0.030927  | 0.018881   | 0.010481   | -0.008370 | -0.002908  | 0.219438    | 0.081759  | 0.088177   | 0.174379   | 0.084002  | 0.070483       | 0.178722   | 0.088177     | 1.00000            | 0.027595           | 0.011844    | 0.081188     | 0.009993     | 0.127480  | 0.004407   | 0.007801  | 0.000990  | 0.116287      |
| TabImplementations | 0.030883      | 0.060793  | -0.008807  | -0.001089  | 0.008231  | -0.000809  | 0.000380    | 0.000320  | 0.000084   | 0.000084   | 0.000084  | 0.000020       | 0.000020   | 0.000020     | 0.000020           | 1.00000            | -0.018885   | -0.000000    | 0.000000     | 0.000000  | 0.000000   | 0.000000  | 0.000000  | 0.000000      |
| Kickstarted        | 0.024078      | 0.128238  | -0.077102  | -0.008895  | 0.016889  | -0.048823  | 0.169685    | 0.380778  | 0.178722   | 0.310871   | 0.004421  | 0.008158       | 0.118084   | -0.022827    | -0.005438          | -0.013117          | 1.00000     | -0.022187    | -0.008895    | -0.001883 | 0.003389   | -0.001883 | 0.048582  | 0.147485      |
| Cat:Thematic       | 0.035100      | 0.290435  | -0.008895  | -0.141236  | 0.001089  | -0.004788  | -0.001089   | -0.010481 | 0.088352   | -0.008762  | -0.013228 | -0.013228      | -0.013228  | -0.013228    | -0.013228          | -0.013228          | -0.013228   | 1.00000      | -0.000000    | -0.000000 | -0.000000  | -0.000000 | -0.000000 | -0.000000     |
| Cat:Strategy       | 0.021242      | 0.444825  | -0.128081  | -0.096384  | 0.018910  | -0.004788  | -0.001089   | -0.010481 | 0.088352   | -0.008762  | -0.013228 | -0.013228      | -0.013228  | -0.013228    | -0.013228          | -0.013228          | -0.013228   | -0.000000    | 1.00000      | -0.000000 | -0.000000  | -0.000000 | -0.000000 | -0.000000     |
| Cat:War            | 0.019435      | -0.128081 | 0.001089   | -0.018910  | -0.010481 | 0.000320   | -0.000320   | -0.000320 | -0.000320  | -0.000320  | -0.000320 | -0.000320      | -0.000320  | -0.000320    | -0.000320          | -0.000320          | -0.000320   | -0.000000    | -0.000000    | 1.00000   | -0.000000  | -0.000000 | -0.000000 | -0.000000     |
| Cat:Family         | 0.010884      | 0.096384  | -0.001883  | -0.010481  | 0.000320  | -0.000320  | -0.000320   | -0.000320 | -0.000320  | -0.000320  | -0.000320 | -0.000320      | -0.000320  | -0.000320    | -0.000320          | -0.000320          | -0.000320   | -0.000000    | -0.000000    | -0.000000 | 1.00000    | -0.000000 | -0.000000 | -0.000000     |
| Cat:OS             | -0.021100     | -0.128837 | 0.108762   | 0.088352   | -0.008762 | -0.013228  | -0.013228   | -0.013228 | -0.013228  | -0.013228  | -0.013228 | -0.013228      | -0.013228  | -0.013228    | -0.013228          | -0.013228          | -0.013228   | -0.000000    | -0.000000    | -0.000000 | -0.000000  | 1.00000   | -0.000000 | -0.000000     |
| Cat:Party          | -0.028853     | -0.184887 | -0.013228  | -0.008762  | -0.013228 | -0.013228  | -0.013228   | -0.013228 | -0.013228  | -0.013228  | -0.013228 | -0.013228      | -0.013228  | -0.013228    | -0.013228          | -0.013228          | -0.013228   | -0.000000    | -0.000000    | -0.000000 | -0.000000  | -0.000000 | 1.00000   | -0.000000     |

Figure 1.3: Pearson correlation matrix

To check the correlation between the attributes, we created a matrix using the *Pearson correlation*. The matrix highlighted the highest correlation between similar

attributes and a strong correlation between related attributes (for example, between GameWeight–ComWeight, NumOwned–NumUserRatings, NumWant–NumWish, NumWeightsVoted–MfgPlayTime–ComMaxPlayTime). For this reason, some attributes were excluded. The correlation level at which we agreed to remove one of the two attributes was whenever the correlation was greater than 0.90.

The reasons for removing other variables were:

- Description, Name, ImagePath: nominal attributes;
- NumComments: all values of the attribute are equal to 0; therefore, the variable is constant and not informative.
- BGGId: an attribute with a unique value for each record.
- Attributes related to Rank were not found to be semantically relevant for subsequent analyses; moreover, in order to avoid data leakage and redundancy phenomena, ranking categories — which do not represent categories with characteristics independent of the game — were excluded.

## Chapter 2 Clustering

The clustering analysis was conducted using exclusively continuous numerical variables. An initial clustering attempt based on a broader set of numerical variables revealed a poorly interpretable structure. For this reason, a clustering subset was defined, consisting of a reduced set of variables that are weakly correlated with each other and exhibit strong semantic relevance with respect to game characteristics. The selected variables are: *ComWeight*, *ComMinPlaytime*, *ComMaxPlaytime*, *NumUserRatings\_log*, *NumWish\_log*.

These five variables represent fundamental dimensions of a game, enabling the synthesis of aspects related to complexity, duration, and the level of popularity/desirability within the community.

Since the numerical variables exhibit very different scales, Min–Max normalization was applied in order to prevent these differences from distorting the computation of Euclidean distances used by the K-means algorithm.

### 2.1 Centroid-based Clustering Algorithm: K-means

To determine the optimal number of clusters, both the *SSE* using the *Elbow Method* and the Silhouette coefficient were computed.

The Elbow plot shows a sharp decrease in error up to  $k = 3$ –4 after which the curve tends to level off. Based on this behavior, the Silhouette coefficient was evaluated over the range from  $k = 3$  and  $k = 6$ . as shown in Figure 2.1.

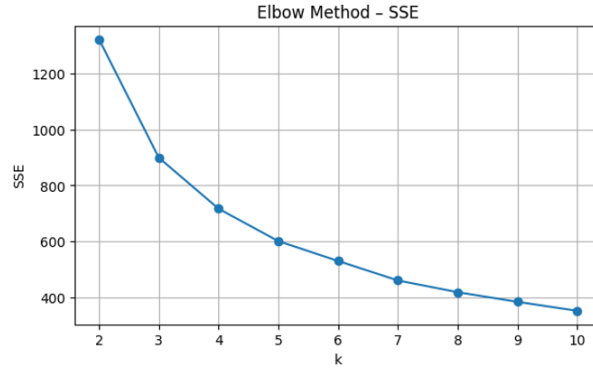


Figure 2.1: Elbow Plot

The highest Silhouette value is obtained for  $k = 3$ , which is therefore the preferred solution. Based on these results,  $k = 3$  was chosen and the model was trained using the *k-means++* initialization, as shown in Figure 2.2.

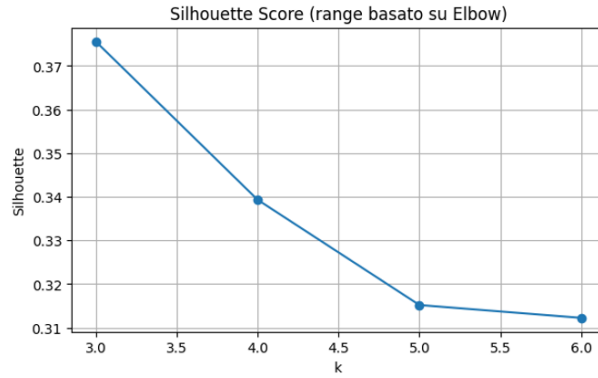


Figure 2.2: Silhouette Score Plot

The application of the model with  $k = 3$  produced an SSE value equal to 898.8931 and a Silhouette coefficient equal to 0.3756, indicating an interpretable cluster structure, albeit with non-sharp separation.

The resulting partition was visualized using a *two-dimensional PCA*, highlighting a clear separation among clusters. Additionally, a *parallel coordinates plot* was employed to compare centroid profiles across the selected variables, as shown in plot 2.4.

The analysis of the centroids allows the identification of three main groups of games, already made evident in plot 2.3:

- Cluster 0: This cluster consists of games with low complexity and relatively short durations. These titles are generally simpler and more accessible, and exhibit lower values in terms of both popularity and desirability within the community.
- Cluster 1: This cluster includes games characterized by high clearly elevated complexity values and very long playtimes, both minimum and maximum. This group encompasses particularly demanding titles, typically associated with extended play sessions and a higher level of difficulty.

- Cluster 2: This cluster comprises games with intermediate levels of complexity and duration, but characterized by high values in the popularity and wishlist variables. This group represents titles that are especially well-received and widely adopted within the community, combining solid gameplay depth with greater immediacy and accessibility.



Figure 2.3: PCA with Clusters

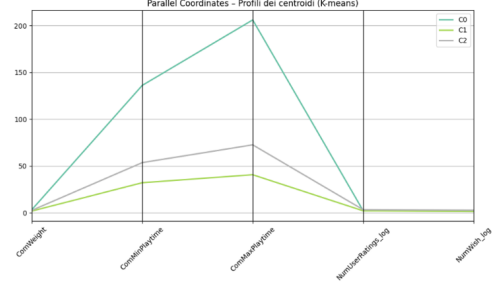


Figure 2.4: Parallel Coordinates

## 2.2 Hierarchical Clustering Algorithm

The hierarchical clustering algorithm was applied using the same variables selected for K-means, after normalization through *Min-Max scaling*. Four linkage criteria were considered: *Single linkage*, *Complete linkage*, *Average linkage* and *Ward linkage*. The resulting dendrograms (2.5) display a strongly unbalanced structure: across all methods, the majority of observations are rapidly merged into a single large group, while only a few points remain separated until the highest levels of the hierarchy. This behavior suggests a high degree of density and continuity in the data distribution, which makes it difficult to identify well-separated clusters.

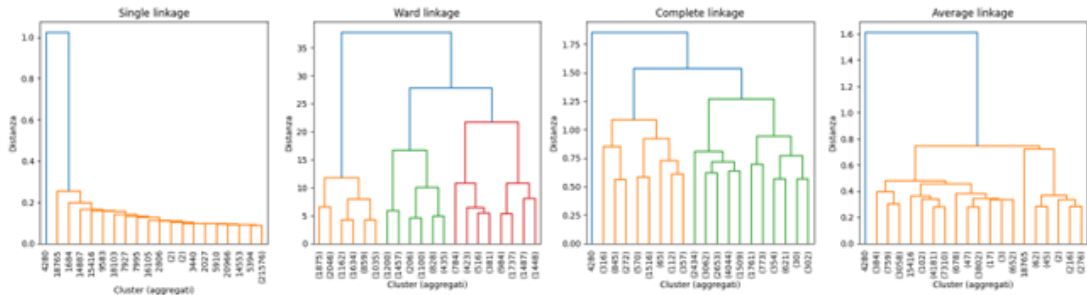


Figure 2.5: Dendrograms

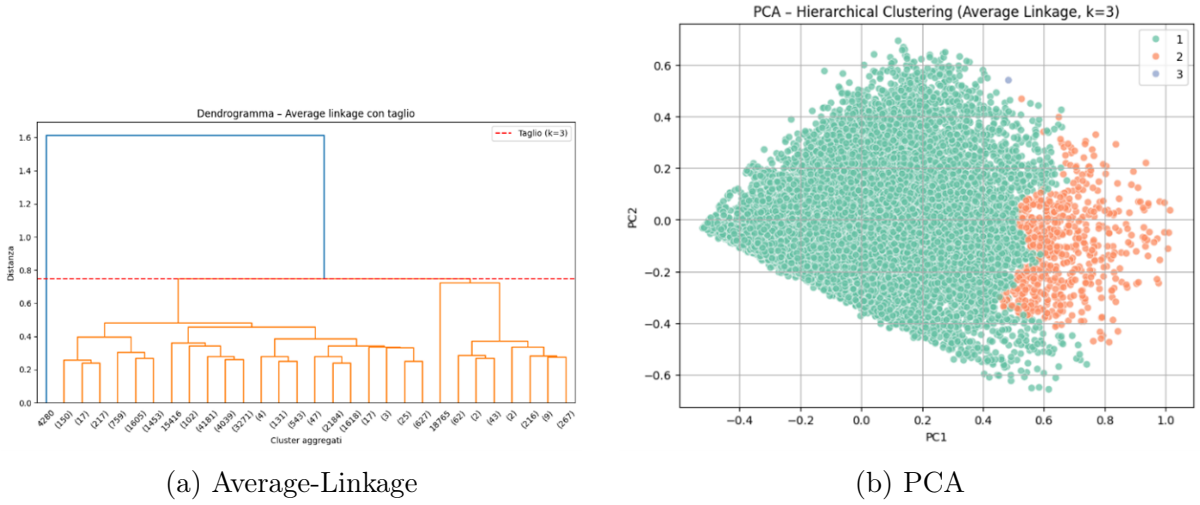
In order to make the results comparable with those obtained using K-means, the dendrogram corresponding to each linkage criterion was cut by imposing  $k = 3$  clusters. The quality of the resulting partitions was assessed using the Silhouette coefficient:

- Single linkage: 0.4955
- Ward linkage: 0.2994

- Complete linkage: 0.3829
- Average linkage: 0.4488

Among the considered methods, *Average linkage* yields the best performance, although the Silhouette value remains moderate, indicating weak or only partial cluster separation.

The final dendrogram obtained with Average linkage and the corresponding cut highlights a strongly unbalanced structure, with one dominant cluster containing the vast majority of observations and two clusters of very small size. The PCA-based visualization confirms this behavior: in the PC1–PC2 plane, the clusters are not clearly separated, and the smaller groups appear as marginal aggregations rather than well-defined, homogeneous sets.



Given the characteristics of the dataset and the selected variables, hierarchical clustering did not yield sufficiently interpretable partitions.

## 2.3 Density-based Clustering Algorithm: DBSCAN

To assess the presence of possible density-based structures, the DBSCAN algorithm was applied to the same variables used for K-means. The parameter *minPts* was fixed at 4, a value commonly adopted for continuous numerical datasets in higher-dimensional spaces.

The second parameter, *eps*, was selected using the k-distance plot, obtained by ordering the distance of each point to its fourth nearest neighbor.

Based on this plot, several values of *eps* within a plausible range were tested, namely: 0.13, 0.10, 0.08 and 0.06.

The obtained results highlight a highly stable behavior:

- With  $\text{eps} = 0.13$ , DBSCAN detects a single large cluster, with only a minimal number of points classified as noise.
- Reducing  $\text{eps}$  to 0.10 and 0.08 again yields a single dominant cluster, accompanied by a slight increase in the number of points identified as outliers.

- With  $\text{eps} = 0.06$ , a few micro-clusters of very small size (on the order of only a few points) emerge, together with one dominant cluster.

The micro-clusters generated for particularly low values of  $\text{eps}$  have negligible cardinalities (from 3 to 14 points), and cannot be interpreted as meaningful clusters. They simply indicate small local irregularities in the data distribution.

Overall, these results suggest that DBSCAN does not identify any meaningful density-based structure within the dataset. The selected variables define a highly compact and uniform feature space, with no natural separations between regions of different density. The selected variables generate a very compact and uniform space, without natural separations between regions with different densities.

Consequently, the application of DBSCAN does not reveal density-based clusters that are useful for data segmentation. In contrast, the centroid-based approach (K-means) proves to be more effective, as it exhibits a clearer and more interpretable clustering structure.

## Chapter 3 Classification and regression

In this section, the first part of the analysis focuses on a supervised classification task, with the aim of predicting board game Ratings—an ordinal categorical variable with three levels (*Low*, *Medium* and *High*)-based on structural and popularity-related features.

The following models were applied and compared: *Decision Tree*, *K-Nearest Neighbors (KNN)*, *Naive Bayes*.

Model performance was evaluated using *Accuracy*, *macro-averaged F1-score*, and *macro-averaged AUC*, with particular attention paid to performance on the individual classes.

### 3.1 Rating Classification – Data Preparation

Before model training, a data preparation phase was carried out.

The target variable Rating was encoded using Label Encoding, transforming the categorical labels (Low, Medium, High) into numerical values.

The following numerical features were selected as independent variables: ComWeight, YearPublished, MinPlayers, MaxPlayers, ComAgeRec, ComMinPlaytime, ComMaxPlaytime, NumUserRatings\_log, NumWish\_log.

The data were split into a training set (75%) and a test set (25%) using a stratified partition, in order to preserve the distribution of the Rating classes in both subsets.

### 3.2 Decision Tree Classifier

#### 3.2.1 DECISION TREE – BASELINE MODEL

The first algorithm considered is the Decision Tree. In an initial phase, a baseline model was trained without imposing constraints on the tree depth, in order to obtain an initial performance benchmark.

The baseline model achieves an Accuracy of 0.588, a macro F1-score of 0.59, and a macro-averaged AUC of 0.686. The confusion matrix highlights a reasonable ability to correctly identify the Low class, while the Medium and High classes are more difficult to discriminate and are frequently confused, consistent with their semantic proximity.

Feature importance analysis shows that the most relevant variables are NumWish\_log, ComWeight, YearPublished, and NumUserRatings\_log, indicating that popularity, perceived complexity, and the recency of the game play a central role in predicting the Rating. Structural variables, such as number of players and game duration, instead provide a more marginal contribution.

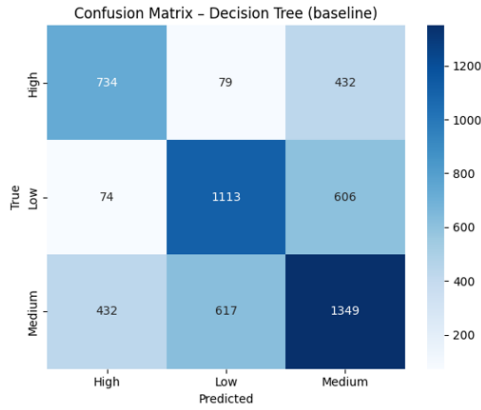


Figure 3.1: Confusion Matrix-Decision Tree

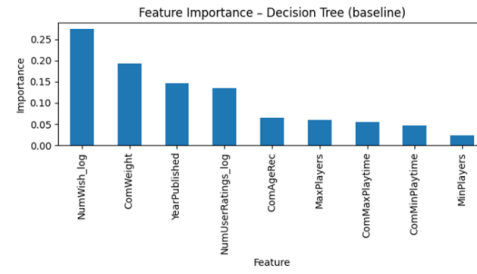


Figure 3.2: Feature Importance-Decision tree

The multiclass ROC curves confirm this behavior, with AUC values around 0.70 for High and Low classes, while the Medium class remains more difficult to discriminate.

### 3.2.2 DECISION TREE – OPTIMIZED MODEL

To improve performance and reduce overfitting, hyperparameter optimization was performed using Grid Search with 5-fold cross-validation, adopting the macro-averaged F1-score as the optimization metric. The optimal configuration identified corresponds to an entropy-based splitting criterion, a maximum tree depth of 10, a minimum of 20 samples per leaf, and a minimum of 2 samples required to perform a split.

The optimized model achieves an Accuracy of 0.670, a macro F1-score of 0.67, and a macro-averaged AUC  $\approx 0.81$ , showing a consistent improvement over the baseline model.

The confusion matrix shows a reduction in overall misclassifications, particularly for the *Low class*. The *High class*, while exhibiting a substantial improvement, remains slightly more challenging to discriminate compared to the other classes.

The multiclass ROC curves highlight a clear improvement, with AUC values of approximately 0.89 for High, 0.87 for Low, and 0.75 for Medium.

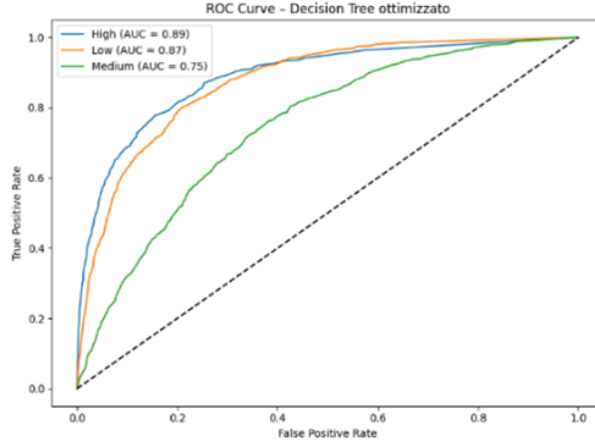


Figure 3.3: Roc Curve - Optimization Decision Tree

Feature importance analysis confirms the central role of NumWish\_log, ComWeight, YearPublished.

Due to the large number of nodes, only the first three levels of the tree are reported for visualization purposes.

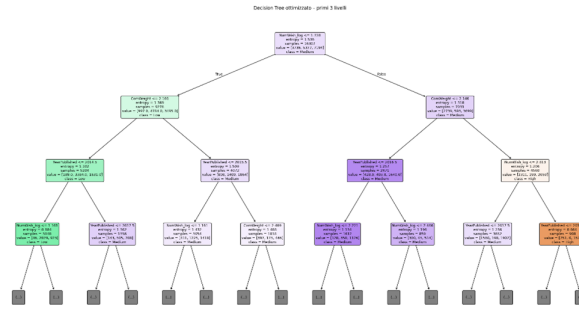


Figure 3.4: Decision Tree Optimization

### 3.3 K-Nearest Neighbors Classifier

The second classification algorithm considered is K-Nearest Neighbors (KNN). Since the method is based on the computation of distances between observations, all numerical variables were first standardized using StandardScaler in order to ensure a homogeneous contribution of the features.

#### 3.3.1 KNN BASELINE

The second classification algorithm considered is K-Nearest Neighbors (KNN). Since the method is based on the computation of distances between observations, all numerical variables were first standardized using StandardScaler in order to ensure a homogeneous contribution of the features.

The baseline model was trained using  $k=5$ , Euclidean distance, and uniform weights. The resulting performance is superior to that of the Decision Tree baseline, achieving an accuracy of 0.65 and a macro F1-score of 0.65.

The analysis of the classification report highlights a good ability to classify the High

and Low classes, while the Medium class remains the most challenging to distinguish, consistently with what was observed for the other models.

The multiclass ROC curves confirm this behavior, with AUC values of 0.90 for the High class, 0.88 for Low, and 0.75 for Medium.

### 3.3.2 OPTIMIZED KNN

To further improve model performance, a hyperparameter optimization phase was carried out using Grid Search with 5-fold cross-validation. The search was implemented through a pipeline, including feature standardization within the validation process, in order to avoid data leakage. The hyperparameter grid explored different values of the number of neighbors, various distance metrics, and alternative weighting schemes.

The best hyperparameters identified are:

- number of neighbors: 15
- distance metric: Manhattan
- weights: uniform

The optimized KNN model achieves an Accuracy of 0.68 and a macro F1-score of 0.68, resulting in the best overall performance in terms of Accuracy and macro F1-score among the analyzed models.

The confusion matrix of the optimized model shows a reduction in misclassifications compared to the baseline, particularly for the Medium class.

The multiclass ROC curves highlight a further improvement in discriminative capability, with AUC values of 0.86 for both the High and Low classes and 0.71 for Medium, confirming the higher effectiveness of the optimized model.

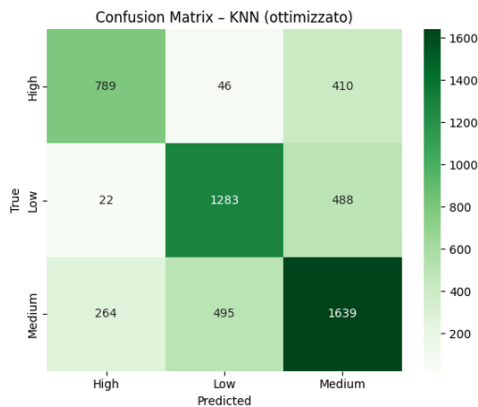


Figure 3.5: Confusion Matrix - Optimization KNN

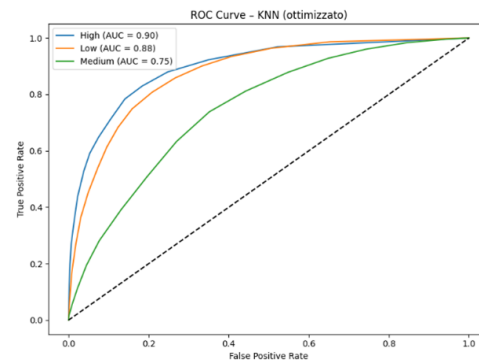


Figure 3.6: ROC Curve - Optimization KNN

## 3.4 Naive Bayes Classifier

The third algorithm considered is Naive Bayes, implemented in its GaussianNB variant, which is suitable for continuous numerical features. The model was trained on the same features used for the other classifiers, after data standardization. The

obtained performance is lower than that of Decision Tree and KNN, with an Accuracy of 0.49 and a macro F1-score of 0.44. The analysis of the confusion matrix highlights a marked difficulty in recognizing the Medium class, which is frequently confused with the Low class. The multiclass ROC curves nonetheless show a good probabilistic discriminative capability for the High and Low classes ( $AUC \approx 0.84$ ), while the Medium class is more difficult to separate ( $AUC \approx 0.66$ ).

The macro-averaged AUC of 0.78 confirms an overall moderate discriminative ability, albeit inferior to that of the more complex models.

Overall, the performance of Naive Bayes appears limited, likely due to the strong violation of the feature independence assumption, making it less suitable for this specific classification task.

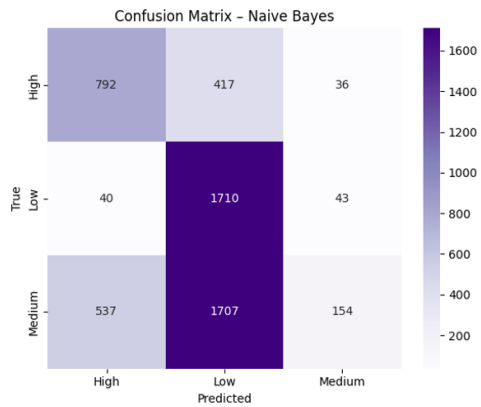


Figure 3.7: Confusion Matrix - Naive Bayes

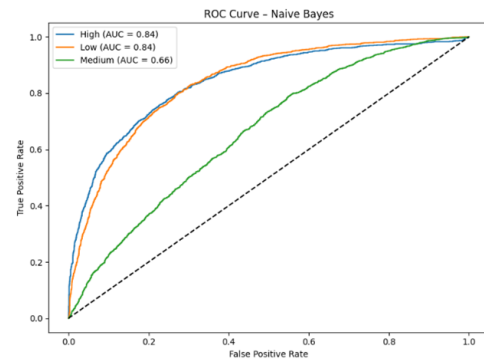


Figure 3.8: ROC curve - Naive Bayes

### 3.4.1 Conclusion

The comparison among the models shows that the optimized KNN achieves the best overall performance in terms of Accuracy, macro F1-score, and macro-averaged AUC. The optimized Decision Tree exhibits very similar results, with a slight reduction in performance but greater interpretability, making it particularly suitable for analyzing the factors influencing the Rating.

| Modello                     | Accuracy | F1 macro | AUC macro |
|-----------------------------|----------|----------|-----------|
| Decision Tree (baseline)    | 0.588    | 0.591    | 0.687     |
| Decision Tree (ottimizzato) | 0.670    | 0.672    | 0.836     |
| KNN (baseline)              | 0.645    | 0.648    | 0.810     |
| KNN (ottimizzato)           | 0.683    | 0.685    | 0.842     |
| Naive Bayes                 | 0.489    | 0.444    | 0.781     |

## 3.5 Regression

In this section, the relationships between different variables were analyzed using supervised regression models, both in univariate and multivariate settings.

Specifically, a linear regression model was adopted with the aim of assessing how the

variable ComWeight is influenced by board game characteristics. For the multivariate regression, nine independent variables were considered, including information related to the recommended age, game duration, level of community engagement, and several structural features of the game.

The dataset was split into a training set (75%) and a test set (25%). Model performance was evaluated using the coefficient of determination ( $R^2$ ) and the mean squared error (MSE); for the univariate regression model, the mean absolute error (MAE) was also considered.

### 3.5.1 UNIVARIATE REGRESSION

A univariate linear regression was tested using ComAgeRec. The model achieved the following performance on the test set:  $R^2 = 0.323$ ,  $MSE = 0.423$ ,  $MAE = 0.498$ . These results indicate the presence of a positive linear relationship between the recommended age and the perceived complexity of the game, although with a high dispersion of the data.

The relatively low  $R^2$  value suggests that a single variable is not sufficient to accurately explain game complexity.

For comparative purposes, Ridge and Lasso regression models were also applied.

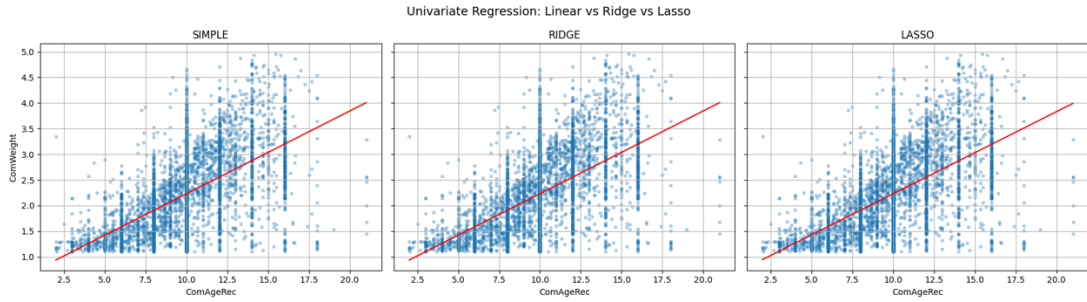


Figure 3.9: Univariate Regression: Linear vs Ridge vs Lasso

### 3.5.2 MULTIVARIATE LINEAR REGRESSION

Multivariate linear regression was performed, considering a set of structural and popularity-related variables of the game.

The model achieves an  $R^2 \approx 0.45$ , showing a clear improvement compared to the univariate case.

This result confirms that game complexity cannot be explained by a single variable, but emerges from the interaction of multiple factors.

Regularized Ridge and Lasso regression models were also considered.

Ridge regression shows no significant improvement over the standard linear model, suggesting limited multicollinearity. Lasso regression slightly improves performance by reducing the impact of less informative variables.

### 3.5.3 NON-LINEAR REGRESSION MODELS

To capture potential non-linear relationships among the variables, two non-linear models were applied. These were K-Nearest Neighbors (KNN) and a Decision Tree Regressor.

| Model  | R <sup>2</sup> | MSE      |
|--------|----------------|----------|
| Linear | 0.447275       | 0.345321 |
| Ridge  | 0.447274       | 0.345321 |
| Lasso  | 0.447800       | 0.344992 |

Table 3.1: Performance comparison of regression models.

- KNN model: was trained on standardized features and optimized with respect to the number of neighbors. The optimized model ( $k = 15$ ) achieves an  $R^2 \approx 0.59$ , showing a substantial improvement over the linear models.
- Decision Tree Regressor: was regularized by limiting its depth to avoid overfitting. The optimal maximum depth is 7. This model achieves an  $R^2 \approx 0.61$ , the highest among those considered.

Feature analysis highlights ComMaxPlaytime as the most important variable. It is followed by ComAgeRec and indicators of community interest. The decision rules show that maximum game duration is central to perceived game complexity.

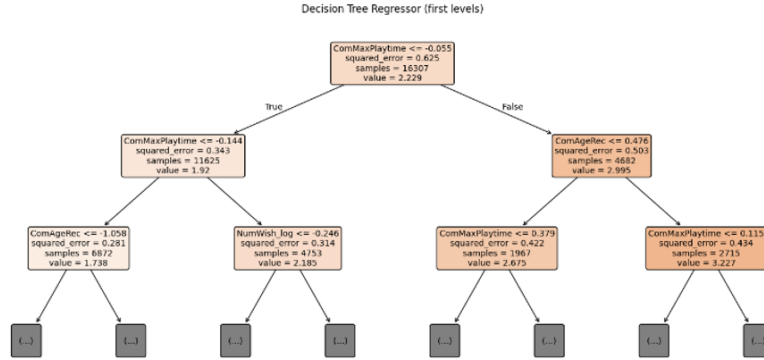


Figure 3.10: Decision Tree regressor

In order to systematically compare the performance of the different regression models considered, Table 3.2 reports the  $R^2$  and MSE values obtained by the models:

| Model         | R <sup>2</sup> | MSE      |
|---------------|----------------|----------|
| Linear        | 0.447275       | 0.345321 |
| Ridge         | 0.447274       | 0.345321 |
| Lasso         | 0.447800       | 0.344992 |
| KNN           | 0.594691       | 0.253221 |
| Decision Tree | 0.605188       | 0.246663 |

Table 3.2: Model performance comparison

# Chapter 4 Pattern mining

In this chapter, a subset of the dataset attributes—previously analyzed in the *data quality* section—is used with the aim of identifying meaningful patterns.

Most numerical variables were discretized using *qcut*, with some exceptions using three bins for popularity measures, in order to avoid unbalanced distributions and poorly informative bins. The Rating attribute, already categorical, was kept unchanged. For some variables, manual or binary intervals were adopted to obtain simpler and more interpretable classes.

## 4.1 Frequent patterns extraction

For the extraction of frequent patterns, the trend in the number of closed and maximal itemsets was analyzed as the minimum support ( $\text{min\_sup}$ ) varied, while fixing the minimum itemset size to  $\text{z\_min} = 2$ . As expected, lower support values lead to a large number of patterns, whereas increasing the threshold progressively reduces their number, eventually stabilizing for higher support levels. Based on this behavior, an intermediate support value was selected to filter out less relevant patterns while retaining the most informative ones and limiting the complexity of the analysis. Closed itemsets preserve support-related information, whereas maximal itemsets provide a more compact representation of frequent patterns. The support threshold was therefore chosen by jointly considering both types of itemsets. The following figure illustrates the variation in the number of closed and maximal itemsets as a function of the minimum support.

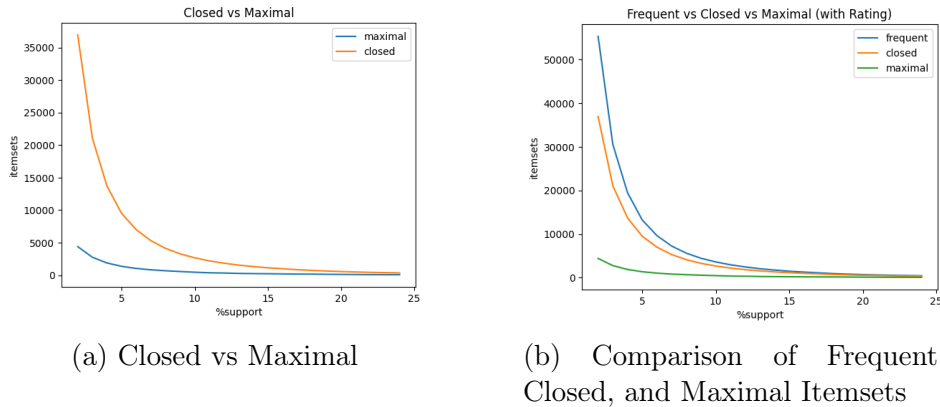


Figure 4.1a shows the trend in the number of closed and maximal itemsets as the minimum support varies, highlighting a marked reduction in the number of extracted patterns as  $\text{min\_sup}$  increases. Figure 4.1b compares frequent, closed, and maximal itemsets, showing that, for higher support values, the three types progressively converge due to the overall decrease in pattern cardinality.

A similar analysis was performed on maximal itemsets containing the Rating attribute, selected as representative of the quality perceived by users. The use of

maximal itemsets is motivated by their greater compactness, which reduces analytical complexity while retaining the most representative feature combinations without introducing redundancy. Also in this case, increasing the minimum support leads to a significant reduction in the number of patterns; in particular, for support values between 15-25%, a convergence is observed due to the limited number of extracted patterns.

a convergence is observed due to the limited number of extracted itemsets.

Based on these observations, an intermediate support threshold of  $\text{min\_sup} = 10\%$  was selected as a compromise between reducing pattern cardinality and preserving a manageable and informative set of itemsets for subsequent analyses.

## 4.2 Association rules

In this section, the most interesting association rules are analyzed, characterized by a lift value greater than 1 and by adequate levels of confidence and support, and obtained using association rule mining algorithms. Different confidence thresholds (30%, 50%, 70%), were tested; among these, the 50% value proved to be the most suitable, as it allows for the extraction of a sufficiently large yet manageable set of rules.

For the extraction of association rules, the same parameters adopted in the previous frequent pattern mining phase were used, fixing  $\text{min\_sup}=10\%$ ,  $\text{zmin}=2$ .

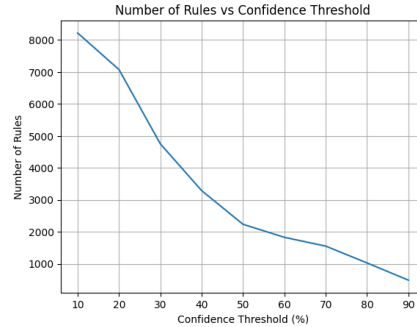


Figure 4.2: Number of rules vs confidence Threshold

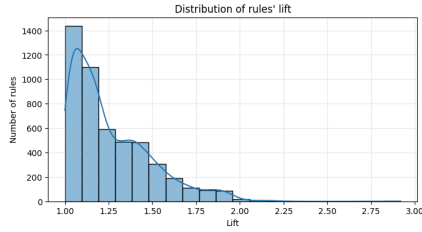
Figure 4.2, 2 was obtained after filtering out rules with a lift value lower than 1. From the figure, a strong reduction in the number of rules can be observed in the confidence range between 20% and 40%, while beyond 80% the decrease becomes less pronounced, as most rules have already been eliminated. The 60–70% range represents a good compromise between rule quality and quantity, but it leads to a significant reduction in the overall number of rules. For this reason, a confidence threshold of 50% was selected, as it allows potentially informative patterns to be preserved while delegating the elimination of less significant associations to the lift measure.

|    | consequent                 | antecedent                                                                             | abs_support | %_support | confidence | lift     |
|----|----------------------------|----------------------------------------------------------------------------------------|-------------|-----------|------------|----------|
| 21 | Is_Reimplementation_Reimpl | (HasImplementations, 10s+_Year)                                                        | 1697        | 7.804811  | 0.660311   | 5.661336 |
| 93 | Kids(<8)_MfgAge            | (Kids(<8)_ComAge, Rapid(<30)_ComMaxTime, Not_Kickstarted_Kickstarter)                  | 1399        | 6.434255  | 0.632745   | 5.004650 |
| 37 | Is_Reimplementation_Reimpl | (HasImplementations, PreTeen(8-13)_ComAge, 0_BestPlayers)                              | 1564        | 7.193120  | 0.574578   | 4.926277 |
| 95 | Kids(<8)_MfgAge            | (Kids(<8)_ComAge, Rapid(<30)_ComMaxTime, 0_BestPlayers)                                | 1406        | 6.466449  | 0.621025   | 4.911946 |
| 34 | Is_Reimplementation_Reimpl | (HasImplementations, PreTeen(8-13)_ComAge, Not_Kickstarted_Kickstarter, 0_BestPlayers) | 1341        | 6.167502  | 0.569669   | 4.884190 |

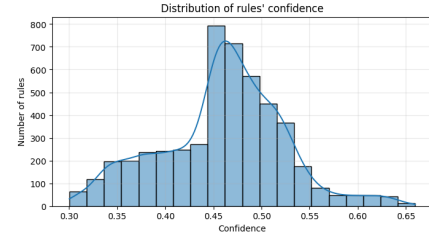
Figure 4.3: Regole di associazione più rilevanti ottenute dall'algoritmo Apriori

Table 4.3 reports the top five rules ranked by decreasing lift. The selected rules are characterized by high values of both lift and confidence, indicating statistically significant associations between antecedent and consequent.

The bar chart in Figure 4.4a shows that, as the lift value increases, the number of rules decreases, with a strong concentration around the value of 1. This behavior is indicative of weak or near-independent associations and is partly attributable to the data discretization process. Only a limited number of rules exhibit truly significant associations, which are concentrated at higher lift values. After applying the filter  $\text{lift} > 1$ , the distribution of confidence values is concentrated in the range between 0.45 and 0.50, suggesting a good balance between the number and the reliability of the extracted rules.



(a) ) Lift Distribution of the Rules



(b) Confidence Distribution of the Rules

For low confidence values, the number of rules is high, as the analysis initially considers all extracted rules without applying any lift-based filtering. The Medium Rating and Low Rating classes exhibit a larger number of rules in the initial phases, while the Low Rating class shows a rapid decrease already beyond the 10% confidence threshold. As confidence increases, the number of rules decreases for all classes and tends to stabilize in the range between 60% and 70%. The High Rating class represents an exception, dropping to zero already at around 50% confidence. These results are reported in Figure 4.5

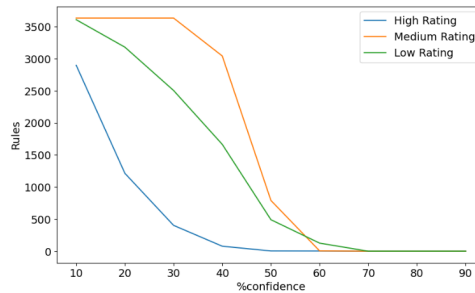


Figure 4.5: Number of association rules as a function of confidence for each Rating class

### 4.3 Target variable prediction

The extracted association rules were used to predict the target variable Rating, which is characterized by a strong prevalence of the Medium Rating class in the training set. After the mining and pruning phases, 60 hybrid rules based on lift and support values were selected and subsequently employed in a weighted voting system that assigns greater weight to more informative rules. The rule-based model shows a moderate predictive capability. An accuracy of 50% and a macro F1-score of 0.49 indicate that the extracted rules are able to capture meaningful relationships within the data, particularly for the High Rating and Low Rating classes. However, the difficulty in correctly discriminating the Medium Rating class penalizes the overall performance of the model, as also reflected by the lower value of the weighted F1-score. Table 4.1 reports the precision, recall, and F1-score values for each class of the target variable Rating.

From the analysis of the confusion matrix 4.6, it can be observed that the model tends to correctly distinguish the Low Rating class, while it shows greater difficulty in recognizing the Medium Rating class, which is frequently confused with the High Rating class. This behavior is consistent with the imbalanced class distribution and helps to explain the overall performance of the model.

Table 4.1: Performance of the rule-based model for each Rating class

| Classe        | Precision | Recall | F1-score |
|---------------|-----------|--------|----------|
| High_Rating   | 0.45      | 0.63   | 0.52     |
| Low_Rating    | 0.51      | 0.77   | 0.62     |
| Medium_Rating | 0.58      | 0.23   | 0.33     |
| Macro Avg     | 0.51      | 0.54   | 0.49     |
| Weighted Avg  | 0.52      | 0.50   | 0.47     |

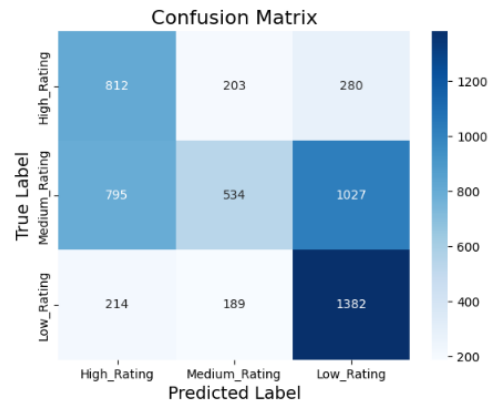


Figure 4.6: Confusion matrix of the rule-based model

From an implementation perspective, both the Apriori and FP-Growth algorithms were employed for the extraction of frequent patterns and association rules. However, as also emerged from the experiments conducted in the notebook, FP-Growth demonstrated superior computational performance, proving to be more efficient in terms of execution time.